MAlang Lexical Analyzer

DFA Minimization Algorithm

CS4031-Compiler Construction | Assignment 01 | Spring 2026

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Objective

To optimize the lexical analyzer by reducing the number of states in the automaton while maintaining the exact same language recognition capabilities.

Implementation Steps

1. **Removal of Null (λ) Transitions:**
   - The initial Non-Deterministic Finite Automaton (NFA) contained null transitions (moves that occur without input).
   - These were eliminated by computing the ε-closure for each state, ensuring every transition is strictly driven by a valid input symbol.
2. **Merging Equivalent States (Repetitive Inputs):**
   - We identified groups of states that exhibited identical behavior, reacting to the same input symbols by transitioning to the same set of subsequent states.
   - These equivalent states were merged. For example, if multiple states handled the same inputs (like different digit states) and led to the same outcome, they were combined into a single representative state.
3. **Removal of Useless States:**
   - The automaton was analyzed to identify "useless" or "dead" states, states that either could not be reached from the start state or could never lead to a final accepting state.
   - These states were omitted entirely from the final transition table, resulting in a compact, minimal Deterministic Finite Automaton (DFA).

Result

The minimization process successfully reduced the state complexity, resulting in the final dfa.csv transition table used by the Manual Scanner.